

The protective effects of escitalopram on cognitive behavior in triple-transgenic mice of Alzheimer's disease

RC Liu¹, ZY Wang², JJ Wang¹, ZJ Wang³, J Zhang³, MN Wu^{3*}, JS Qi^{3*}



¹ The Second Hospital of Shanxi Medical University, Taiyuan, China; ² The First Hospital of Shanxi Medical University, Taiyuan, China; ³ Department of Physiology, Key Laboratory of Cellular Physiology, Ministry of Education, Shanxi Medical University, Taiyuan, China

BACKGROUND

Alzheimer's disease (AD) is a neurodegenerative disease characterized by progressive memory decline and cognitive impairment. Depressive symptoms occur commonly in the course of AD. The concept of 'depression-dementia medius' was proposed in clinical practice. AD has various pathological hallmarks, including cerebral amyloid senile plaques, neurofibrillary tangles, synaptic and neuronal loss in the brain. In addition, studies believe that decreased serotonin levels in the post mortem frontal cortex is a putative marker of accelerated cognitive decline in AD. Escitalopram, a selective serotonin reuptake inhibitors (SSRIs) antidepressant, which ameliorated serotonin decrease in both depression and AD, is used to relieve depression among AD patients. Moreover, for patients with mild cognitive impairment, a prodromal state of AD, SSRIs could improve the memory and cognitive function. Previous studies have shown that SSRIs are able to inhibit amyloid- β production and amyloid plaques in transgenic animal models and AD patients. SSRIs increase neurogenesis in the dentate gyrus of hippocampus. However, whether escitalopram can ameliorate memory deficit in the AD mice was still unknown.

MATERIALS & METHODS

Animals	3xTg-AD mice		C57BL/6J mice	
	Female, 9 months		Female, 9 months	
Drugs	Escitalopram (i.g.)	Saline (i.g.)	Escitalopram (i.g.)	Saline (i.g.)
	10 mg/kg·d 12weeks	same volume, 12 weeks	10 mg/kg·d 12 weeks	same volume, 12weeks
Behavioral Tests	Elevated plus-maze		Tail suspension test	
	Y-maze		Morris water maze	

RESULTS

1. Escitalopram ameliorated depression-like behavior and behavioral disinhibition of 3xTg-AD mice

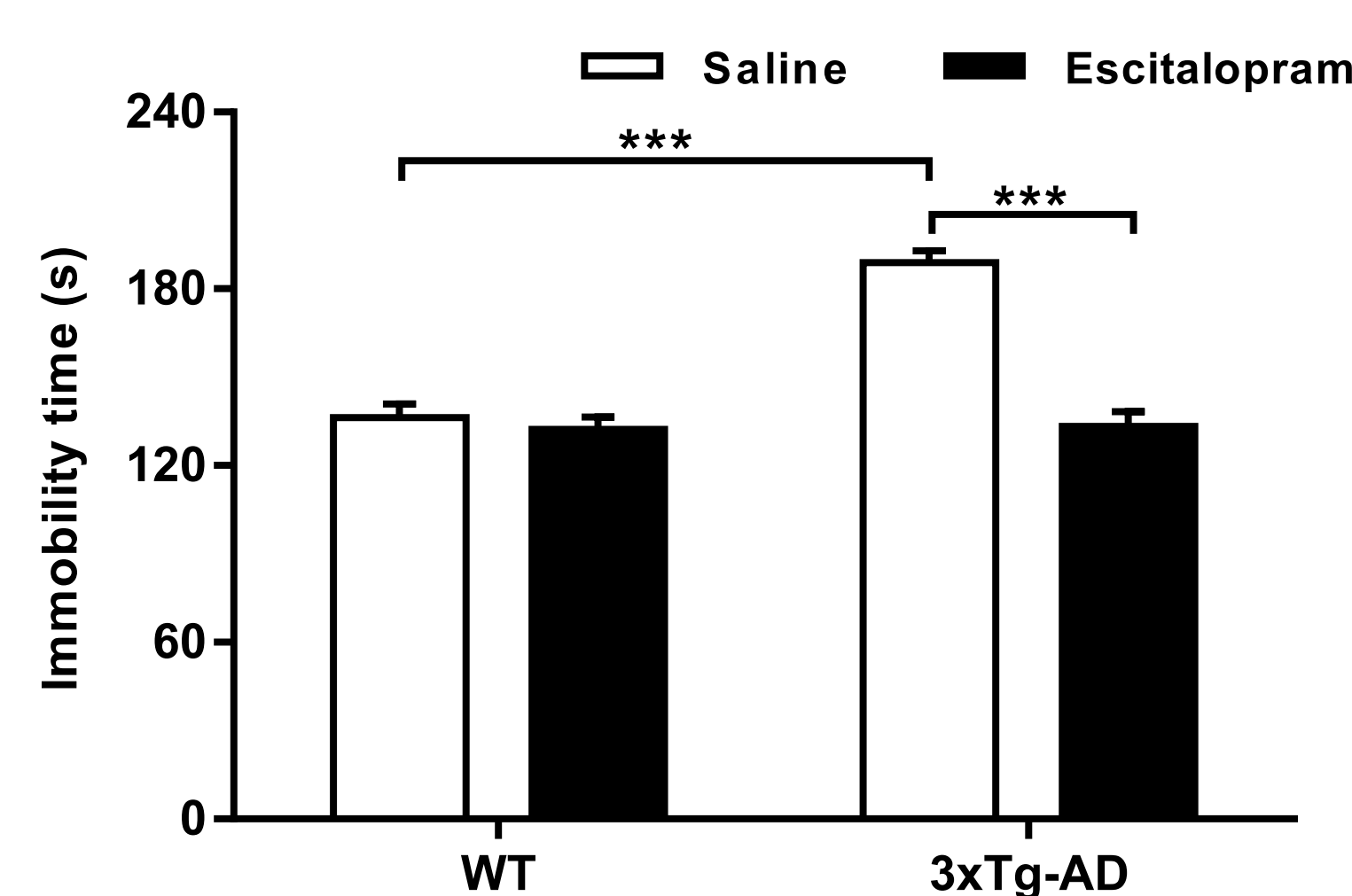


Fig. 1. Escitalopram decreased immobility time of 3xTg-AD mice in the tail suspension test. Each column represents the mean $\pm$ S.E.M. ($n = 10-11/\text{group}$, *** $P < 0.001$).

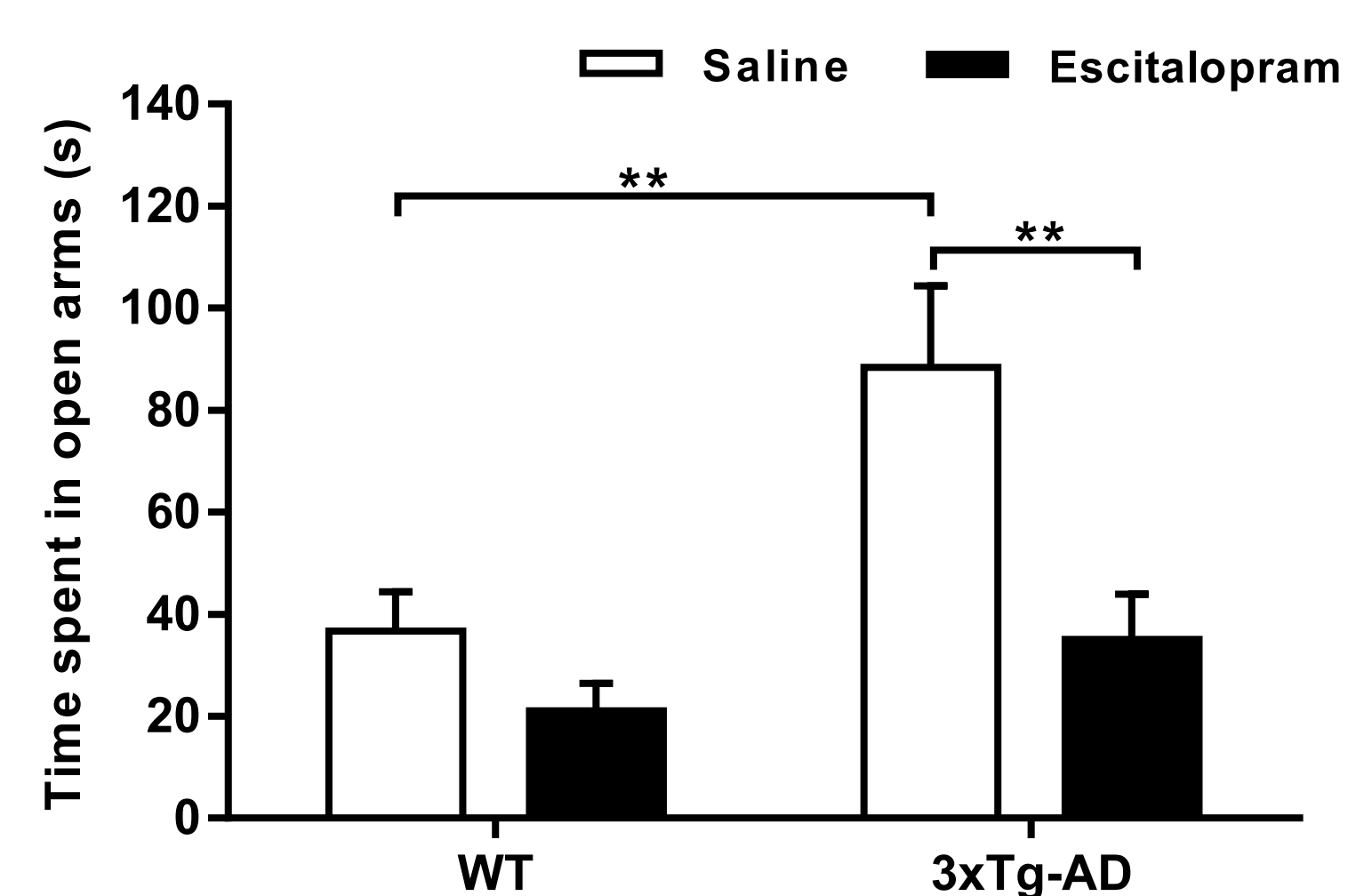


Fig. 2. 3xTg-AD mice showed behavioral disinhibition in the elevated plus-maze test at the age of 12 months, which were ameliorated by escitalopram. Each column represents the mean $\pm$ S.E.M. ($n = 10-11/\text{group}$, ** $P < 0.01$).

2. Escitalopram improved the working memory of 3xTg-AD mice

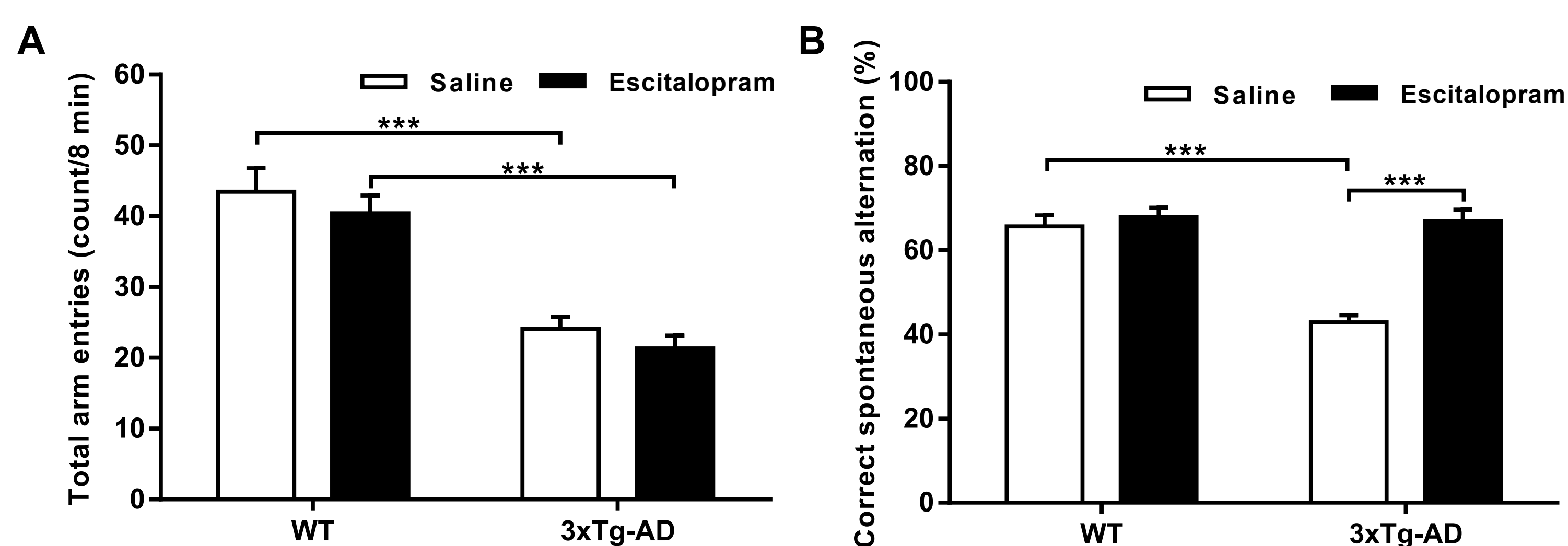


Fig. 3. Escitalopram improved working memory of 3xTg-AD mice in the Y-maze test. **A** 3xTg-AD mice showed a significant decrease in total arm entries of mice in 8 min compared with WT mice, which were not affected by escitalopram. **B** The percentage of correct spontaneous alternation in 3xTg-AD+Saline group was lower than other group. Each column represents the mean $\pm$ S.E.M. ($n = 10-11/\text{group}$, *** $P < 0.001$).

3. Escitalopram alleviated the spatial cognition deficits of 3xTg-AD mice

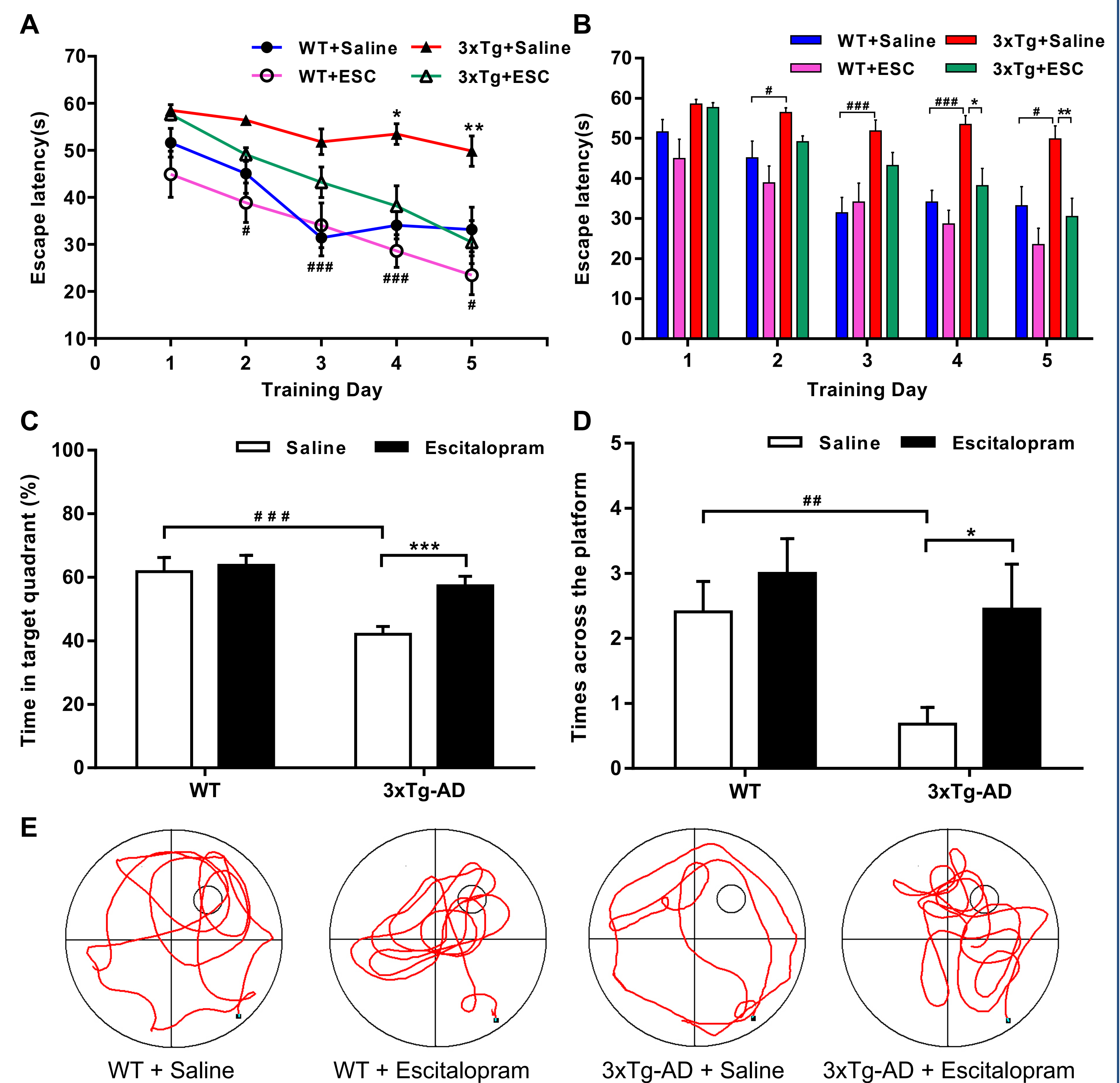


Fig. 4. Effects of escitalopram on spatial learning and memory of 3xTg-AD mice detected by the Morris water maze test. **A** and **B** Average escape latencies of mice in acquisition trial phase in 5-day training trials. The escape latency of 3xTg-AD mice was longer than that in WT mice from day 2 to day 5, and escitalopram reversed it at last two days. **C** and **D** Percentage of swimming time in target quadrant (**C**) and times across the platform (**D**) in the probe trial phase. The time in target quadrant and times across the platform in 3xTg-AD + Saline group was lower than that in WT + Saline group, and escitalopram treatment effectively ameliorated it. Each column represents the mean $\pm$ S.E.M. ($n = 10-11/\text{group}$; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, 3xTg-AD group vs. WT group; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, escitalopram treated 3xTg-AD vs. controlled 3xTg-AD group). **E** Representative swimming tracks in the probe trials. (large circles, water maze pool; small circles, platform; black dots, starting points; $n = 10-11/\text{group}$).

CONCLUSIONS

In this study, we tested the effects of escitalopram on behavioral performance in 12-month-old female 3xTg-AD mice. Chronic administration with escitalopram effectively ameliorated behavioral disinhibition and depression-like behavior of AD mice. Escitalopram treatments also improved short-term and long-term spatial learning and memory of 12-month-old AD mice. The present study provides a new evidence for the neuroprotection of antidepressant drugs, suggesting that escitalopram, may be beneficial for the treatment of AD. Further clinical trials will be necessary.

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